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Manifold-based trajectory planning for safe proximity and soft docking to a tumbling target by modular multi-spacecraft system

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ABSTRACT

This paper proposes a manifold-based trajectory planning method to achieve safe proximity to a tumbling target by a modular multi-spacecraft system. This method solves the optimal control problem under boundary value constraints, collision avoidance constraints, and soft docking constraints by introducing the mathematical concept of differential manifolds. First, the state manifold space of the modular multi-spacecraft system is constructed, and a potential function is introduced to build the manifold structure. Second, an innovative homotopy class construction method based on rotating 3D-Dubins curves is proposed. Compared with traditional homotopy initial guesses, this method can make the manifold-based method converge from an initial guess that does not satisfy the dynamics constraints to an optimized result with soft docking performance. Finally, through simulation comparisons with the model predictive control-sequential convex programming method and semi-physical simulations based on an air flotation experimental system, the advantages of the proposed method in terms of energy optimization, soft docking, and computational efficiency are verified.

1. Introduction

With the rapid development of space technology, modular multi-spacecraft systems have shown increasingly important application value in the fields of on-orbit servicing and space assembly [1–3]. Especially in mission scenarios such as collaborative handling and target maintenance, the modular spacecraft system needs to achieve close-range safe proximity to the target [4]. This process faces two key challenges: First, it is necessary to coordinate the movement of multiple modules to achieve energy optimization. Second, it is necessary to overcome the safety risks caused by the target's failure-induced tumbling motion. Among them, accurately satisfying the collision avoidance constraint is the primary condition to ensure the safety of the approach process [5–7].

In order to increase the safety of the approach process, the modular spacecraft needs to achieve soft docking with the target in addition to being collision-free. Soft docking is defined as the relative terminal velocities and positions of the two spacecraft being kept within tolerances [8–10]. The terminal radial velocity should be as small as possible to ensure the safety of both the module and the target. Therefore, the essence of using a modular multi-spacecraft system to achieve safe proximity to a tumbling target is to solve an optimal control problem under the constraints of boundary value, collision avoidance, and soft docking.

However, traditional trajectory methods face numerous challenges in addressing such problems. Typically, optimal control problems are primarily solved using indirect and direct methods [11]. The indirect method is based on the optimal control method of Pontryagin's minimum principle [12]. By introducing the costates and Hamiltonian function, the optimal control problem is transformed into solving the two-point boundary value problem (TPBVP). The direct method transforms the continuous optimal control problem into a nonlinear programming problem and solves it using numerical methods, such as sequential convex optimization [13]. The indirect method is more accurate than the direct method but has the disadvantages of relying on an initial costate guess and having a smaller domain of convergence. The direct method has a larger domain of convergence but does not guarantee optimality [14].

Therefore, traditional trajectory planning faces three major challenges. The first is the inherent sensitivity to initial values, which requires a good initial guess that conforms to the dynamic constraints to achieve algorithm convergence [15]. Second, for the control problem of

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modular multi-spacecraft systems, these methods suffer from the problem of increased system dimensionality due to multiple constraints [16]. The added constraints also complicate the problem, for example, by requiring the introduction of costates or the convexification of the constraints. Finally, traditional solution methods typically focus on studying the dynamics model of each spacecraft individually [17,18]. This approach may overlook the dynamic characteristics and interactions of the multi-spacecraft system as a whole, failing to fully leverage the synergies within the system and thereby compromising the effectiveness of overall energy optimization.

Developing a new trajectory planning method that accurately describes the overall dynamics of the multi-spacecraft system, reduces reliance on initial guesses, and does not increase the number of optimization variables is crucial for improving the efficiency and safety of multi-spacecraft system missions. Differential manifolds, as a class of research objects in differential geometry, are widely used to solve complex dynamic control problems [19]. Radhakrishnan [20] pointed out that the traditional trajectory planning algorithm makes the system prone to falling into local optima, essentially because the state space is a manifold characterized by global non-Euclidean properties and high dimensionality. By studying the multi-spacecraft system motion and control problem in low-dimensional manifolds, the proliferation of system dimensions can be effectively avoided [21]. Belabbas and Liu [22,23] described the state space of affine nonlinear systems, such as unmanned vehicles and robots, as a manifold. The solution of the two-point boundary value problem is found by searching the geodesic on the manifold. The process of searching for geodesics is realized by homotopy iteration. This manifold-based method transforms the two-point boundary value problem of optimal control into the problem of finding geodesics on manifolds. Sipowa [24] applied this method to the spacecraft trajectory planning problem and proved that the spacecraft minimum fuel consumption trajectory planning problem is equivalent to solving geodesics on manifolds.

In summary, compared to Euclidean space approaches, the manifold-based method offers four key advantages: (1) Reduced dependence on initial guesses. It converges reliably from simple initial guesses, even those violating dynamic constraints [24]. (2) The likelihood of getting stuck at a local extreme value is reduced. Through homotopy parameterization, it enables transitions between local extrema, particularly beneficial for complex constrained trajectory planning [25]. (3) The introduction of multiple constraints will not increase the system dimension. It intrinsically satisfies constraints via geometric structure [26]. (4) It takes into account the overall dynamic properties of a multi-spacecraft system and can help properly understand the intrinsic dynamics of the system [19].

However, when faced with the mission of achieving a safe approach to a tumbling target using a modular multi-spacecraft system, the current manifold-based approach still suffers from shortcomings. First of all, the current research only considers a single spacecraft when constructing manifold space and does not consider the multi-spacecraft system as a whole [27]. Secondly, the geodesic search process cannot consider the soft docking performance. This is mainly due to the limitations imposed by the construction of the homotopy initial guess for the geodesics. There is a conflict between soft docking and energy optimization. The traditional geodesic solution method usually uses a linear [28] or trigonometric function [24] as the homotopy initial guess. These guesses do not consider the solution with soft docking performance in the iterative process. Iterating from these guesses causes the final optimal approximate solution to not have soft docking properties, since soft docking is treated as a soft constraint in this case. Therefore, the soft docking can be realized by designing a reasonable homotopy initial guess.

The Dubins curve is the shortest path that connects two points in a two-dimensional plane under the constraints of curvature and boundary tangents [29]. Compared to the linear guess, Dubins curves can account for the soft docking requirement. Therefore, Dubins curves can evolve a homology class that satisfies the soft docking constraints [30]. Improving the Dubins curve and using it as an initial guess can enable the manifold-based method to converge to a trajectory optimization result that satisfies the dynamics while also providing soft-docking performance.

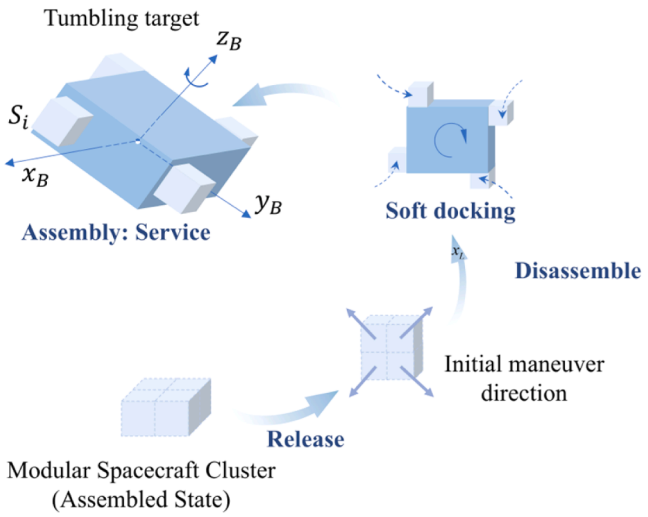


Fig. 1. Illustration of a modular multi-spacecraft system approaching a tumbling target.

In this paper, for the safe proximity trajectory planning problem of a modular multi-spacecraft system approaching a tumbling target, an innovative multi-spacecraft system trajectory planning framework is proposed by introducing the mathematical concept of differential manifolds and validated on an air-float experimental platform. The framework has the following advantages:

(1) Firstly, the multi-spacecraft system is regarded as a whole, and the overall manifold space is constructed. This helps to understand and describe the dynamic behavior of the multi-spacecraft system more comprehensively. Compared with the traditional model predictive control-sequential convex programming (MPC-SCP) method, this approach can obtain a solution with less energy consumption, which is verified in simulations.

(2) Second, the potential function is used to design a manifold metric based on its continuously differentiable nature. The metric is characterized by a clear physical meaning. The design of the manifold structure using this metric yields smooth and continuous velocities. The velocity smoothness is verified by comparison in the simulation section.

(3) Finally, a method for constructing homotopy initial guesses based on rotated 3D-Dubins curves is proposed. On the one hand, this framework can achieve convergence from initial guesses that only satisfy the boundary value constraints, thereby reducing dependency. On the other hand, the framework can converge to solutions with soft-docking properties. The effectiveness of the method has been verified on an air-float experimental platform.

The paper is organized as follows. Section 2 models the scenario and the optimization problem. Section 3 proposes a multi-spacecraft system trajectory planning method based on manifolds. Section 4 proposes a homotopy class construction method based on rotated 3D-Dubins curves as an initial guess for the manifold-based method. Section 5 proposes a manifold metric construction method based on potential functions. Section 6 presents simulation and experimental platform validation results, showing the advantages of the proposed method in energy optimization and soft docking constraint handling by comparing it with the classical algorithm MPC-SCP. Additionally, the optimization process of the algorithm for energy is analyzed through simulation. Section 7

summarizes the whole paper.

2. Problem formulation

The mission scenario of this paper is the on-orbit servicing of a tumbling target using a modular multi-spacecraft system. The research objective is to achieve trajectory planning and energy optimization for the multi-spacecraft system. In particular, the terminal states need to satisfy soft docking constraints. Soft docking is defined as the terminal velocity and position remaining within tolerance, where the terminal radial velocity is kept as small as possible to ensure the safety of modules and targets [9]. Finally, on-orbit assembly scenarios generally require all modules to arrive at the capture point simultaneously.

As shown in Fig. 1, the initial state of the modular multi-spacecraft system is an assembled configuration. The multi-spacecraft system acquires an initial maneuvering velocity for module release. The released and disassembled modules each follow a planned trajectory to achieve soft docking with the target capture point.

Before constructing the modular multi-spacecraft system trajectory optimization model, the dynamics model is first constructed.

2.1. Dynamic model construction

Three coordinate frames are introduced for the tumbling motion of the target as well as the relative motion of the multi-spacecraft system:

(1) LVLH coordinate frame $\{\mathscr{L}\}$: located at the center of mass of the target, $+x_L$ axis along the radial direction, $+y_L$ axis along the orbital direction, and $+z_L$ axis along the orbital momentum moment.
(2) Target body-fixed frame $\{\mathscr{B}\}$: located at the center of mass of the target spacecraft, with $+x_B$, $+y_B$, and $+z_B$directed along principle body axes.
(3) Inertial coordinate frame $\{\mathscr{I}\}$: located at the Earth's center, with the $+z_I$ axis aligned along the Earth's rotational axis and pointing toward the North Pole, the $+x_I$ axis pointing toward the vernal equinox, and the $+y_I$ axis completing a right-handed coordinate system with the $+x_I$ and $+z_I$ axes.

Considering the close-range safe proximity scenario, the target spacecraft moves in a circular orbit. The spacecraft in the multi-spacecraft system follows the CW relative motion. The multi-spacecraft system can be expressed as the following affine system:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \tag{1}$$

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}, \mathbf{A}_{ij} = \begin{bmatrix} \widehat{\mathbf{A}}_{ij} & & \mathbf{O} \\ & \ddots & \\ \mathbf{O} & & \widehat{\mathbf{A}}_{ij} \end{bmatrix}_{3N\times 3N}, \mathbf{B} = \begin{bmatrix} \mathbf{0}_{3N\times 3N} \\ \mathbf{I}_{3N\times 3N} \end{bmatrix} \tag{2}$$

$$\widehat{\mathbf{A}}_{11} = \mathbf{0}_{3\times 3}, \widehat{\mathbf{A}}_{12} = \mathbf{I}_{3\times 3}, \widehat{\mathbf{A}}_{21} = \begin{bmatrix} 3n^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -n^2 \end{bmatrix}, \widehat{\mathbf{A}}_{22} = \begin{bmatrix} 0 & 2n & 0 \\ -2n & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{3}$$

where N is the number of spacecraft in the multi-spacecraft system, $\mathbf{x} = [\underbrace{\boldsymbol{\rho}}_{3N\times 1}^{\mathrm{T}} \ \underbrace{\mathbf{v}}_{3N\times 1}^{\mathrm{T}}]^{\mathrm{T}}$ represents the state of all spacecraft, $\mathbf{u} = [\underbrace{\mathbf{u}_1^{\mathrm{T}} \ \cdots \ \mathbf{u}_N^{\mathrm{T}}}_{3N\times 1}]^{\mathrm{T}}$ denotes the control input, and n signifies the mean motion of the target orbit.

The target spacecraft performs uncontrolled tumbling motion. Suppose that the position vectors of a capture point on the target in the frames $\{\mathscr{L}\}$ and $\{\mathscr{B}\}$ are $\boldsymbol{\rho}_L^c$ and$\boldsymbol{\rho}_B^c$, respectively, and the transformation relationship between them is:

$$\boldsymbol{\rho}_L^c = \mathbf{T}_B^L \boldsymbol{\rho}_B^c \tag{4}$$

where $\mathbf{T}_B^L$ is the transition matrix from $\{\mathscr{B}\}$ to $\{\mathscr{L}\}$. $\mathbf{T}_B^L$ is key to analyzing the instantaneous state of the target capture point and can be directly determined from the transformation relationship between coordinate frames. As shown in Eq. (5), the transition matrix from $\{\mathscr{I}\}$ to $\{\mathscr{L}\}$ is represented using Euler angles:

$$\mathbf{T}_I^L = \mathbf{T}_z(f)\mathbf{T}_x(inc)\mathbf{T}_z(\Omega) \tag{5}$$

where Ω is the right ascension of ascending node, inc is the orbital inclination, and f is the trueanomaly. The transition matrix from $\{\mathscr{I}\}$ to $\{\mathscr{B}\}$ is:

$$\mathbf{T}_I^B = \mathbf{T}_z(\phi)\mathbf{T}_x(\theta)\mathbf{T}_z(\psi) \tag{6}$$

where ψ is the precession angle, θ is the nutation angle, and ϕ is the rotation angle. Let the target's rotational angular velocity be $\boldsymbol{\omega}_B = [\omega_{Bx}, \omega_{By}, \omega_{Bz}]^{\mathrm{T}}$, the moment of inertia be $\mathbf{J} = \mathrm{diag}([J_x, J_y, J_z])$, $J_x = J_y = J_e$, $J_z = J_p$, and the angular momentum be H. Then the calculation formulas for ψ, θ, and φ are as follows:

$$\begin{cases} \psi = \psi_0 + \omega_e t \\ \theta = \arccos(J_p\omega_p/H) \\ \phi = \phi_0 + \omega_r t \end{cases} \tag{7}$$

ψ_0 and ϕ_0 represent the initial attitude of the scene. Therefore, $\mathbf{T}_B^L$ is:

$$\mathbf{T}_B^L = \mathbf{T}_I^L(\mathbf{T}_I^B)^{\mathrm{T}} \tag{8}$$

Let the rotational angular velocities of the frames $\{\mathscr{L}\}$ be $\boldsymbol{\omega}_L = [0, 0, n]^{\mathrm{T}}$, then the relative angular velocity between frame $\{\mathscr{L}\}$ and frame $\{\mathscr{B}\}$ is:

$$\boldsymbol{\omega} = \mathbf{T}_B^L\boldsymbol{\omega}_B - \boldsymbol{\omega}_L \tag{9}$$

In summary, converting the Euler angle representation to quaternion form and setting the rotation quaternion from $\{\mathscr{B}\}$ to $\{\mathscr{L}\}$ as $\mathbf{q} = [q_0 \ \ q_1 \ \ q_2 \ \ q_3]^{\mathrm{T}}$, the six-degree-of-freedom kinematic and dynamic model for the capture point is [31]:

(1) Kinematics:

$$\dot{\rho}_L^c = \boldsymbol{\omega} \times \boldsymbol{\rho}_L^c \tag{10}$$

$$\mathbf{q} = \frac{1}{2}\mathrm{mati}(\mathbf{q})\mathbf{T}_L^B\boldsymbol{\omega} \tag{11}$$

(2) Dynamics:

$$\ddot{\rho}_L^c = \dot{\omega} \times \boldsymbol{\rho}_L^c + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{\rho}_L^c) \tag{12}$$

$$\dot{\omega} = \mathbf{T}_B^L\dot{\omega}_B + \boldsymbol{\omega} \times (\mathbf{T}_B^L\boldsymbol{\omega}_B) - \dot{\omega}_L \tag{13}$$

$$\mathbf{J}\dot{\omega}_B + \boldsymbol{\omega}_B \times \mathbf{J}\boldsymbol{\omega}_B = 0 \tag{14}$$

where $\mathrm{mati}(\mathbf{q})$ is the local metamorphic matrix of quaternion $\mathbf{q}$:

$$\mathrm{mati}(\mathbf{q}) = \begin{bmatrix} -q_1 & -q_2 & -q_3 \\ q_0 & -q_3 & q_2 \\ q_3 & q_0 & -q_1 \\ -q_2 & q_1 & q_0 \end{bmatrix} \tag{15}$$

2.2. Multi-spacecraft system trajectory optimization model

The trajectory optimization model of the module multi-spacecraft system can be defined as a two-point boundary value problem:

- Objective: energy optimal for the multi-spacecraft system
- Constraints:

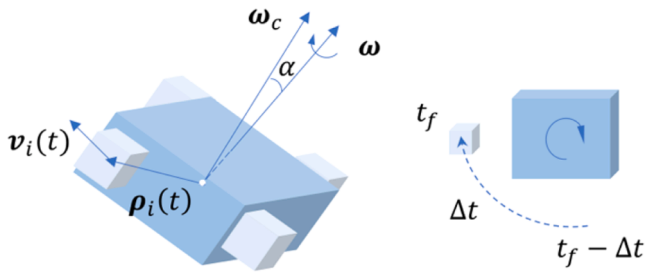


Fig. 2. Illustration of soft docking performance.

(1) Dynamics constraint. The multi-spacecraft system satisfies the dynamics of Eq. (1).

(2) Collision avoidance constraints. Firstly, collision avoidance needs to be realized within the multi-spacecraft system:

$$\Delta\rho = \| \boldsymbol{\rho}_i - \boldsymbol{\rho}_j \|_2 \geq R_c, i, j \in N, i < j \tag{16}$$

Secondly, the modules cannot collide with the target:

$$\| \boldsymbol{\rho}_i \|_2 \geq R_t, i \in N \tag{17}$$

where $\boldsymbol{\rho}_i$ is the position vector of the i-th spacecraft, R_c is the minimum collision avoidance radius between spacecraft within the multi-spacecraft system, and R_t is the minimum collision avoidance radius between the multi-spacecraft system and the target.

(3) Boundary value constraints. The initial state constraints are:

$$\boldsymbol{\rho}_i(t_0) = \boldsymbol{\rho}_{i0}, \mathbf{v}_i(t_0) = \mathbf{v}_{i0} \tag{18}$$

Let the position vector of the target capture point of the i-th module in the frame $\{\mathscr{B}\}$ be $\boldsymbol{\rho}_B^{ic}$. According to Eqs.(4) and (10), the terminal soft docking constraint is:

$$\boldsymbol{\rho}_i(t_f) = \mathbf{T}_B^L \boldsymbol{\rho}_B^{ic}, \mathbf{v}_i(t_f) = \boldsymbol{\omega} \times \left(\mathbf{T}_B^L \boldsymbol{\rho}_B^{ic}\right) \tag{19}$$

(4) Time constraint. The mission time is T, and the multi-spacecraft system needs to arrive at the capture point simultaneously at the end of the mission:

$$t_f = T \tag{20}$$

In summary, the modular multi-spacecraft system trajectory optimization model is:

$$\begin{cases} \min J = \sum_{i=1}^{N} \int_{t_0}^{t_f} \| \mathbf{u}_i \|_2 \mathrm{d}t = \int_{t_0}^{t_f} \| \mathbf{u} \|_2 \mathrm{d}t \\ \text{s.t.} \\ \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \Delta\rho = \| \boldsymbol{\rho}_i - \boldsymbol{\rho}_j \|_2 \geq R_c, i \in N \\ \| \boldsymbol{\rho}_i \|_2 \geq R_t, i \in N \\ \boldsymbol{\rho}_i(t_0) = \boldsymbol{\rho}_{i0}, \mathbf{v}_i(t_0) = \mathbf{v}_{i0} \\ \boldsymbol{\rho}_i(t_f) = \mathbf{T}_B^L \boldsymbol{\rho}_B^{ic}, \mathbf{v}_i(t_f) = \boldsymbol{\omega} \times \left(\mathbf{T}_B^L \boldsymbol{\rho}_B^{ic}\right) \\ t_f = T \end{cases} \tag{21}$$

To evaluate the soft-docking performance of the planned trajectory, as illustrated in Fig. 2, the angle $\alpha(t)$ is defined. This angle represents the difference between the instantaneous angular velocity $\boldsymbol{\omega}_c(t) = \boldsymbol{\rho}_i(t) \times \mathbf{v}_i(t)$ of each module in the multi-spacecraft system and the target angular velocity $\boldsymbol{\omega}(t)$. The time domain before the end of the scenario is denoted as Δt. To enhance safety and achieve better soft-docking properties, the variation of $\alpha(t)$within Δt should be minimized and ideally take the value of 0 at the terminal moment.

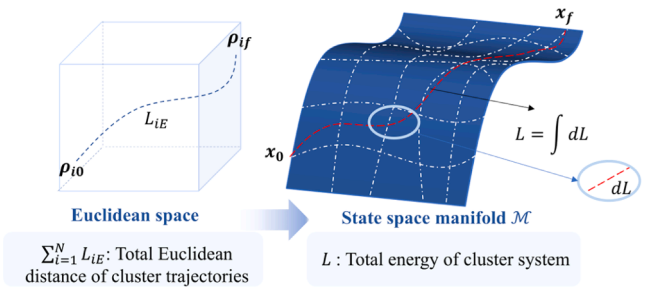


Fig. 3. Comparison of manifold space and Euclidean space.

3. Manifold-based multi-spacecraft system trajectory planning method

In this section, the mathematical concept of differential manifold is introduced, and the two-point boundary value problem of Eq. (21) is transformed into the geodesic search problem on manifold.

3.1. Definition of geodesics

The state space of the modular multi-spacecraft system is actually an abstract curved space called a manifold ($\mathscr{M}$). The solution to Eq. (21) is an integral curve γ on $\mathscr{M}$. As shown in Fig. 3, Euclidean space often evaluates the length of the trajectory by calculating the sum of Euclidean distances ($\sum_{i=1}^{N} L_{iE}$) for the modular spacecraft. However, the length of the trajectory is not directly related to the system's energy. In an abstract manifold such as state space, the length L of the integral curve γ can represent the total energy of the system [32]. The smaller L is, the less total energy the system consumes.

To obtain the optimal solution of Eq. (21), it is necessary to find the integral curve on the manifold $\mathscr{M}$ that minimizes L. The shortest path between two points on the manifold is called the geodesic. To determine the geodesic, one must first define a ruler to measure the length of the parametric curve in the manifold. Let the line element of the parametric curve in Fig. 3 be denoted as dL, defined using the inner product.

$$\mathrm{d}L = \left(\langle \dot{\mathbf{x}}(t), \mathbf{G}\dot{\mathbf{x}}(t) \rangle\right)^{\frac{1}{2}} \tag{22}$$

Let $\mathbf{G}$ be the metric matrix that corrects for the difference between the ruler on the manifold space and the ordinary inner product of Euclidean space. As a correction term, $\mathbf{G}$ can alter the structure of the manifold space. Therefore, with a reasonable design of $\mathbf{G}$, the geodesics in the manifold space can satisfy both the equality and inequality constraints in Eq. (21). Consequently, the geodesic is defined as:

$$\min L = \min \int \left(\langle \dot{\mathbf{x}}(t), \mathbf{G}\dot{\mathbf{x}}(t) \rangle\right)^{\frac{1}{2}} \mathrm{d}t \tag{23}$$

The solution of Eq. (23) implies that the second derivative at a point on the integral curve is zero. In the differential manifold, the second derivative is defined as an operator ∇, known as a connection. The solution of Eq. (23) is thus $\nabla_{\dot{\mathbf{x}}(t)} \dot{\mathbf{x}}(t) = 0$, where:

$$\nabla_i = \partial_i + \Gamma_{ij}^k \tag{24}$$

where the subscripts and superscripts i, j, k are indices of the components of the state vector $\boldsymbol{x}$, following the Einstein summation convention. They are dummy variables that can be rotated, used to denote generalized coordinates on the manifold, and their range of values corresponds to the dimension of the state space. ∂_i denotes the ordinary derivative in Euclidean space, and Γ_{ij}^k is a correction term known as the Christoffel symbol. TheΓ_{ij}^k is calculated as:

$$\Gamma_{ij}^k(\mathbf{x}) = \frac{1}{2} \sum_l g^{kl} \left(\frac{\partial g_{li}}{\partial x_j} + \frac{\partial g_{lj}}{\partial x_i} - \frac{\partial g_{ij}}{\partial x_l} \right) \tag{25}$$

where g_{ij} and g^{il} are the components of the metric matrix $\mathbf{G}$ and its inverse matrix $\mathbf{G}^{-1}$, respectively.

3.2. Solution of geodesics using PDE

The computation of the Christoffel symbol and the treatment of boundary conditions make the direct solution of Eq. (23) challenging. To address this, scholars [22,23] have proposed the introduction of a homotopy concept to find geodesics. The core idea is to define a homotopy initial guess as the initial value of the integral curve, iteratively refining this guess to approach the geodesic. When combined with the system dynamics Eq. (1) for the modular multi-spacecraft system, this iterative process can be expressed by a partial differential equation (PDE):

$$\begin{aligned}\frac{\partial \mathbf{x}(t,s)}{\partial s} &= \frac{\partial(\dot{\mathbf{x}}-\mathbf{A}\mathbf{x})}{\partial t} + \sum_{j,k}(\dot{\mathbf{x}}_j - \mathbf{A}\mathbf{x}_j)\frac{\partial \mathbf{x}_k}{\partial t}\Gamma^i_{jk} + \mathbf{r}(t,s) \\ \mathbf{r}(t,s) &= \mathbf{G}^{-1}\left(\left(\frac{\partial \mathbf{A}\mathbf{x}}{\partial \mathbf{x}}\right)^{\mathrm{T}}\mathbf{G}(\dot{\mathbf{x}}-\mathbf{A}\mathbf{x}) + \frac{1}{2}(\dot{\mathbf{x}}-\mathbf{A}\mathbf{x})\left(\frac{\partial \mathbf{G}}{\partial \mathbf{x}}\right)(\mathbf{A}\mathbf{x})\right)\end{aligned} \tag{26}$$

where, s is a homotopy parameter with a value of $s \in [0, s_{\max}]$. The PDE has been shown to evolve in the direction of energy reduction. s replaces the optimization process of trajectory planning in the physical sense. Therefore, when $s = 0$, the curve $\mathbf{x}(t, 0)$represents the initial guess. When $s = s_{\max}$, the curve $\mathbf{x}(t, s_{\max})$ represents the energy-optimal approximate solution. $\mathbf{r}(t, s)$ is the correction term that corrects the drift motion in Eq. (1). $\mathbf{r}(t, s)$ can compensate for the influence of the drift term $\boldsymbol{Ax}$ on the geodesic, allowing the trajectory to maintain its natural drift characteristics under the affine system [22]. Additionally, since $t \in [0, T]$, the solution of the PDE corresponds to the solution of a two-point boundary value problem at a fixed time.

In summary, the two-point boundary value problem of Eq. (21) is transformed into the PDE solution problem of Eq. (26). This paper will focus on the following two important issues:

(1) Design of the initial guess$\mathbf{x}(t,0)$. It is necessary to design an initial guess that only needs to satisfy the boundary value constraints of Eqs.(18) and (19), without needing to meet the dynamics constraints and collision avoidance constraints. Additionally, the solution obtained from this guess should have good soft-docking performance.
(2) Design of the metric matrix $\mathbf{G}$. The metric matrix $\mathbf{G}$ needs to be constructed to modify the manifold structure, ensuring that the PDE iterates in a direction that satisfies all constraints.

4. Rotated 3D-Dubins curves for initial guess construction

Equation (26) generally uses linear [28] or trigonometric [24] initial guesses:

$$\mathbf{x}(t,0) = (1 - t/T)\mathbf{x}_0 + (t/T)\mathbf{x}_f \tag{27}$$

$$\mathbf{x}(t,0) = \mathbf{x}_0\cos((\pi/2)\cdot(t/T)) + \mathbf{x}_f\sin((\pi/2)\cdot(t/T)) \tag{28}$$

These guesses lack soft-docking performance. Due to the large solution space of Eq. (21), the choice of homotopy class affects the geometric properties of the final trajectory. Therefore, this paper introduces and improves upon the Dubins curve as the initial guess by considering the geometric constraints imposed by soft docking and the characteristic properties of the rotating target dynamics model. Eq. (26) can use this improved guess as the initial solution and iteratively converge to a globally approximate optimal solution with soft-docking performance.

4.1. Analysis of rotating target dynamics

First, the dynamics model of the capture point on the tumbling target, as described in Eqs. (10)-(14), is analyzed. Take the moment of inertia $J = \mathrm{diag}([20.2667, 20.2667, 36.48])\mathrm{kg{\cdot}m^2}$, mass $m = 152$ kg, and rotational velocity $\boldsymbol{\omega}_B = -[4.5, 1.5, 1.5]^{\circ}/\mathrm{s}$. The capture point is $\boldsymbol{\rho}_B^c = [1, 0, 0]^{\mathrm{T}}\mathrm{m}$, and the initial quaternion is $\mathbf{q} = [1, 0, 0, 0]^{\mathrm{T}}$. The trajectory of the capture point in the frame $\{\mathscr{I}\}$ is illustrated in Fig. 4.

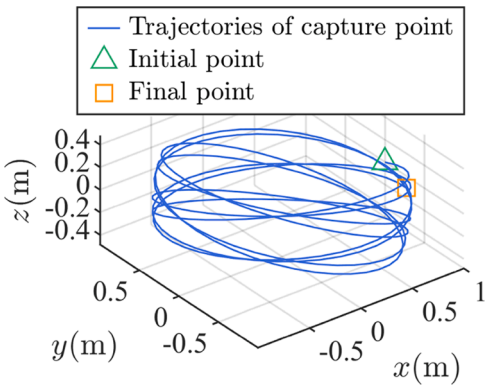


Fig. 4. The trajectory of the capture point.

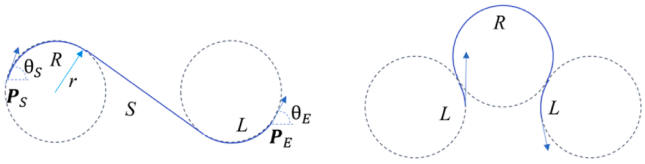


Fig. 5. Composition of 2D-Dubins curves.

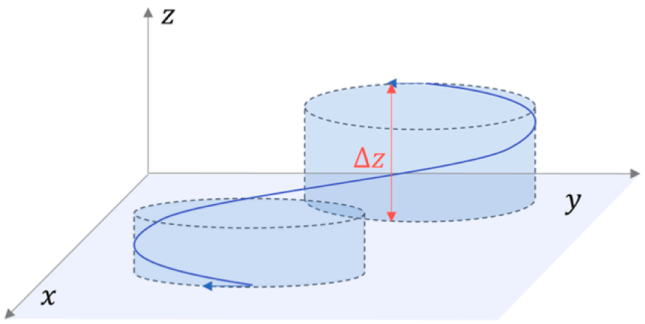


Fig. 6. Solving 3D-Dubins curve by interpolation.

As shown in Fig. 4, the capture point undergoes circular motion with a time-varying rotation plane, necessitating a 3D spiral trajectory for soft docking. While spiral motions (e.g., Archimedean spirals [9]) can be used as rendezvous paths, they present two limitations for our scenario (Fig. 1): (1) failure to incorporate initial maneuvering velocity direction constraints, and (2) restrictive trajectory forms that may yield suboptimal energy performance.

4.2. Traditional Dubins curve

Therefore, in this paper, an initial guess based on rotated 3D-Dubins curves is proposed. A Dubins curve is the shortest path connecting two points in the two-dimensional plane, satisfying curvature constraints and initial/terminal tangent directions. It has the characteristic of a circular spiral at the terminal position and considers the maneuvering direction at the initial position. As shown in Fig. 5, the 2D-Dubins curve is mainly classified into two categories: CSC and CCC. It consists of a series of circular arcs and straight line segments, which can be categorized into $\{RSR, RSL, LSR, LSL, RLR, LRL\}$ according to the left-turn/right-turn arcs. The three-segment curve can be denoted as Segment-1

($\boldsymbol{\rho}_{s1}$),Segment-2($\boldsymbol{\rho}_{s2}$),Segment-3($\boldsymbol{\rho}_{s3}$).

2D-Dubins curve is an established algorithm. It can be solved by knowing the radius of curvature r, the starting point $\mathbf{P}_S$, the ending point $\mathbf{P}_E$, and the corresponding tangent directions θ_S and θ_E. However, for the safe approach motion of a tumbling spacecraft, the 2D-Dubins curve needs to be extended to 3D. As shown in Fig. 6, 3D-Dubins curves are generally realized by linear interpolation or Bessel interpolation in the z-direction [33].

4.3. Rotated 3D-Dubins curves

As shown in Fig. 7, the modular spacecraft's initial maneuver from the assembled to separated state is primarily in-plane to minimize out-of-plane fuel consumption. To achieve the desired terminal spiral motion while accounting for this initial direction, two circular arcs serve as initial guesses. The Dubins curve's radius of curvature is set to $r = \| \boldsymbol{\rho}_f \|_2$, with $\mathbf{v}_0/\| \mathbf{v}_0 \|_2$ and $\mathbf{v}_f/\| \mathbf{v}_f \|_2$ defining the start/end tangent directions, respectively.

The 3D-Dubins curve consists of three planar segments: Segment-1 in the orbital plane, Segment-3 in the instantaneous rotation plane, and connecting Segment-2. The junction points are denoted as Flag1 and Flag2.

As shown in Fig. 7 the rotational quaternions $\mathbf{q}$ from $\{\mathscr{B}\}$ to $\{\mathscr{L}\}$ follow a 3-1-3 Euler sequence ($\xi - \beta - \vartheta$. Segment-1 ($\boldsymbol{\rho}_{2D-s1}$) and Segment-3 ($\boldsymbol{\rho}_{2D-s3}$) are constructed as 2D-Dubins curves in their respective planes, connected via Segment-2 ($\boldsymbol{\rho}_{2D-s2}$) at junctions Flag1 ($t_{Flag1}$) and Flag2 ($t_{Flag2}$). Segment-3 is then rotated by β around the x_L-axis to align with the instantaneous rotation plane.

$$\boldsymbol{\rho}_{3D-s3} = \mathbf{T}_x(\beta)\boldsymbol{\rho}_{2D-s3} \tag{29}$$

where $\mathbf{T}_x(\beta)$ denotes the x_L-axis rotation matrix. According to Eq. (8), $\beta = \arccos(T_{33})$, where T_{33} denotes the (3,3) element of matrix $\mathbf{T}_B^L$. Segment-1 remains in the orbital plane to satisfy the initial maneuver constraint ($\boldsymbol{\rho}_{3D-s1} = \boldsymbol{\rho}_{2D-s1}$). Then Segment-2 ($\boldsymbol{\rho}_{3D-s2}$) is.

$$\boldsymbol{\rho}_{3D-s2}(t) = \boldsymbol{\rho}_{3D-s1}(t_{Flag1}) + \frac{\boldsymbol{\rho}_{3D-s1}(t_{Flag1}) - \boldsymbol{\rho}_{3D-s3}(t_{Flag2})}{\| \boldsymbol{\rho}_{3D-s1}(t_{Flag1}) - \boldsymbol{\rho}_{3D-s3}(t_{Flag2}) \|_2}(t - t_{Flag1}) \tag{30}$$

$\boldsymbol{\rho}_{3D-s1}$, $\boldsymbol{\rho}_{3D-s2}$, and $\boldsymbol{\rho}_{3D-s2}$ together form a rotated 3D-Dubins curve. Since the initial guess for the PDE equations only needs to satisfy the boundary constraints, the simplest linear velocity planning is used as the velocity guess:

$$\mathbf{v}(t,0) = (1 - t/T)\mathbf{v}_0 + t/T\mathbf{v}_f \tag{31}$$

In summary, the initial guess algorithm for the PDE is shown in Algorithm 1.

Iterating the PDE with the initial guess of Algorithm 1 yields an energy-optimal solution that overcomes the limitations of the rotated 3D-Dubins curve and the linear velocity constraint of Eq. (31). The converged solution satisfies all dynamics, boundary conditions, collision avoidance constraints, and soft docking performance, and the iterative process is explained in detail in Section 6.2.

5. Manifold metric construction method based on potential function

Algorithm 1 generates the initial guess $\mathbf{x}(t,0)$ for PDE (Eq. (26)). The metric matrix $\boldsymbol{G}$ modifies the manifold structure to ensure PDE iterations satisfy all constraints, while correction term $\mathbf{r}(t,s)$ compensates for drift motion $\mathbf{Ax}$. As shown in Fig. 8, $\boldsymbol{G}$ guides the solution toward collision-free trajectories by adapting the manifold geometry to obstacle presence.

It is proved in the refs [34,35] that collision avoidance on a Riemannian manifold can be represented by an artificial potential function. Therefore, in this paper, the collision avoidance constraints of Eqs. (16) and (17) are uniformly characterized in the metric matrix $\mathbf{G}$ based on the potential function.

The modular spacecraft is modeled as a cube with side length M, and the target has dimensions$L$, W, and$H$. For the purpose of trajectory planning, both the modular spacecraft and the target are treated as spheres with radii $r_s = \sqrt{2}M/2$and$r_t = \sqrt{L^2 + H^2 + W^2}/2$, respectively, as depicted in Fig. 9. Then the minimum collision avoidance radius in Eqs.(16) and (17) are $R_c = 2r_s$, $R_t = r_t + r_s$, respectively.

Replace the collision avoidance constraints of Eqs.(16) and (17) with the following potential function:

$$V_1(\mathbf{x}) = \begin{cases} K_1 \sum_{j \neq k} \frac{1}{2}\eta_1 \left(\frac{1}{\| \boldsymbol{\rho}_j - \boldsymbol{\rho}_k \|_2} - \frac{1}{R_c} \right)^2 & \text{if} \| \boldsymbol{\rho}_j - \boldsymbol{\rho}_k \|_2 < R_c \\ 0 & \text{else} \end{cases} \tag{32}$$

Algorithm 1
PDE initial guess.

Input: $\boldsymbol{\rho}_f$,$\boldsymbol{\rho}_0$, $\mathbf{v}_f$,$\mathbf{v}_0$, T, t
Result: $\mathbf{x}(t,0)$
for i = 1to N do
Calculate the 2D-Dubins curve for spacecraft i in the orbital plane:
Radius of curvature of Dubin $\leftarrow r = \| \boldsymbol{\rho}_f \|_2$
starting point $\leftarrow \mathbf{P}_S = [\rho_{x0}, \rho_{y0}, 0]$
endpoint $\leftarrow \mathbf{P}_E = [\rho_{xf}, \rho_{yf}, 0]$
Tangent direction of the starting point $\leftarrow \theta_S = \arccos\left(v_{x0}/\sqrt{v_{x0}^2 + v_{y0}^2}\right)$
Tangent direction of the endpoint $\leftarrow \theta_E = \arccos\left(v_{xf}/\sqrt{v_{xf}^2 + v_{yf}^2}\right)$
($\boldsymbol{\rho}_{2D-s1}$,$\boldsymbol{\rho}_{2D-s2}$,$\boldsymbol{\rho}_{2D-s3}$)$\leftarrow$2D-Dubins($r$,$\mathbf{P}_S$,$\mathbf{P}_E$,$\theta_S$,$\theta_E$)
Computing the times of the connection points t_{Flag1},t_{Flag2}
$\boldsymbol{\rho}_{3D-s1}(t) = \boldsymbol{\rho}_{2D-s1}(t)$
if $t > t_{Flag2}$ then
$\boldsymbol{\rho}_{3D-s3}(t) = \mathbf{T}_x(\beta)\boldsymbol{\rho}_{2D-s3}(t)$
else if $t_{Flag1} < t < t_{Flag2}$ then
$\boldsymbol{\rho}_{3D-s2}(t) \leftarrow$ Eq. (30)
end if
$\boldsymbol{\rho}_i(t,0) = [\boldsymbol{\rho}_{3D-s1}(t)^{\mathrm{T}}, \boldsymbol{\rho}_{3D-s2}(t)^{\mathrm{T}}, \boldsymbol{\rho}_{3D-s3}(t)^{\mathrm{T}}]^{\mathrm{T}}$
$\mathbf{v}_i(t,0) \leftarrow$ Eq. (31)
end for
$\mathbf{x}(t,0) = [\boldsymbol{\rho}_1(t,0)^{\mathrm{T}}, ..., \boldsymbol{\rho}_N(t,0)^{\mathrm{T}}, \mathbf{v}_1(t,0)^{\mathrm{T}}, ..., \mathbf{v}_N(t,0)^{\mathrm{T}}]^{\mathrm{T}}$

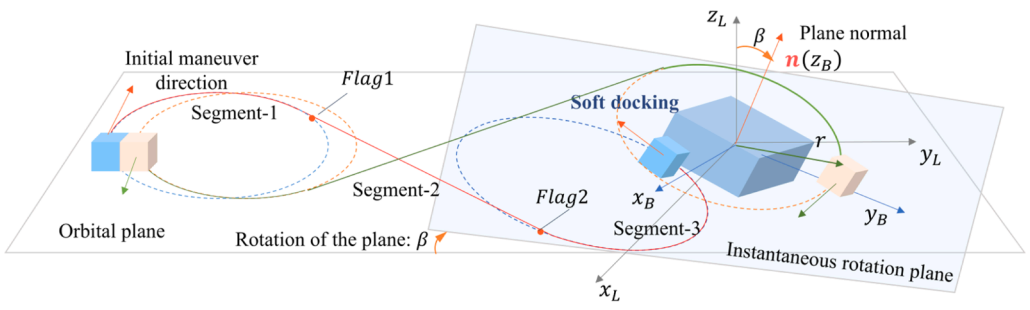


Fig. 7. Illustration of rotated 3D-Dubins curves.

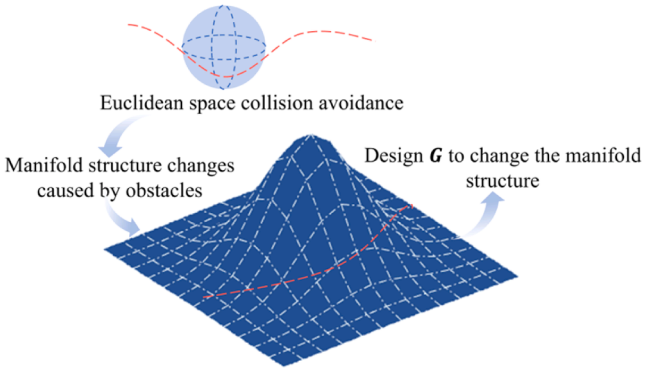


Fig. 8. Relationship between collision avoidance constraints and manifold metrics.

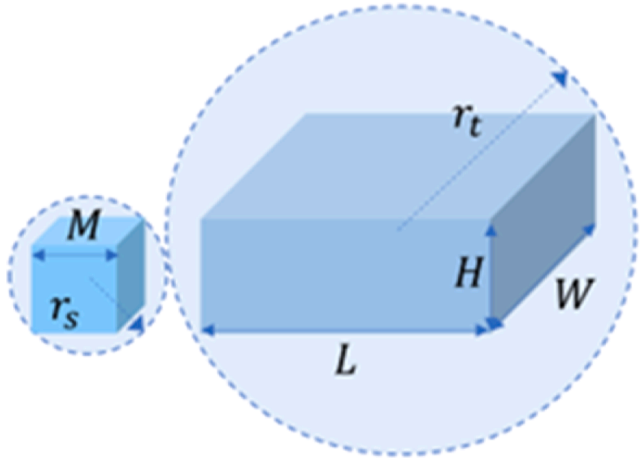


Fig. 9. Illustration of minimum collision avoidance radius.

Algorithm 2
The solution process of the PDE.

Input: $\boldsymbol{\rho}_f$,$\boldsymbol{\rho}_0$, $\mathbf{v}_f$,$\mathbf{v}_0$, T,$s_{\max}$
Result: $\mathbf{x}(t,s_{\max})$
Determine the domain of PDE: $x(t,s):[0,T]\times[0,s_{\max}]$
Calculate the initial guess of PDE: Algorithm 1
Compute metric matrix G←Eq. (34)
Compute Christoffel symbols Γ_{ij}^{k}←Eq. (25)
Solve PDE Eq. (26)

$$V_2(\mathbf{x})=\begin{cases} K_2\sum_{i=1}^{N}\frac{1}{2}\eta_2\left(\frac{1}{\|\boldsymbol{\rho}_i\|_2}-\frac{1}{R_t}\right)^2 & \text{if}\|\boldsymbol{\rho}_i\|_2<R_t \\ 0 & \text{else}\end{cases} \tag{33}$$

where, K_1 and K_2 are scaling factors introduced to prevent the potential function from becoming pathological. η_1 and η_2 are potential function repulsion factors used to determine the magnitude of the constraint action. Therefore, the design metric matrix $\mathbf{G}$ is:

$$\mathbf{G}(\mathbf{x})=(1+V_1(\mathbf{x})+V_2(\mathbf{x}))\mathbf{I}_{6N\times 6N} \tag{34}$$

Eq. (34) guarantees that the metric matrix is symmetric as well as derivable. In summary, the PDE solution problem of Eq. (26) can be organized as Algorithm 2.

6. Simulations and experiments

This section validates the manifold-based multi-spacecraft system trajectory planning method. Firstly, it is compared with the MPC-SCP [16] algorithm to verify the advantages of low energy consumption and soft docking of the proposed method in this paper. The convergence effect of the initial guess is further analyzed. Finally, the effectiveness of the method is verified by the air-float experiment platform.

Table 1
Scenario parameters.

Parameter	Value
Orbital parameters	Circular orbit with orbital altitude $h=726$ km Ω=0, inc=0, f =0
Shape parameters of module spacecraft	$M=0.4$m
Shape parameters of target	$L=3M, W=3M, H=M$
Moment of inertia of the target	$J=\text{diag}([20.2667, 20.2667, 36.48])\text{kg}\cdot\text{m}^2$
Initial attitude quaternion of the target	$\mathbf{q}=[1,0,0,0]^{\mathrm{T}}(\psi_0=0,\phi_0=0)$
The angular velocity of the frame $\{\mathscr{B}\}$	$\boldsymbol{\omega}_B=[-0.0013,-0.0013,-0.0039]$rad /s
The angular velocity of the frame $\{\mathscr{L}\}$	$\boldsymbol{\omega}_L=[0,0,0.0011]^{\mathrm{T}}$rad /s
Number of modular multi-spacecraft systems N	N=4
The initial state of the multi-spacecraft system in frame $\{\mathscr{L}\}$	S_1 $\boldsymbol{\rho}_{10}=[-10.2838;0.2838;0]$m,$\mathbf{v}_{10}=[0;0.01;0]$m /s S_2 $\boldsymbol{\rho}_{20}=[-9.7162;0.2838;0]$m,$\mathbf{v}_{20}=[0.01;0;0]$m /s S_3 $\boldsymbol{\rho}_{30}=[-10.2838;-0.2838;0]$m,$\mathbf{v}_{30}=[-0.01;0;0]$m /s S_4 $\boldsymbol{\rho}_{40}=[-9.7162;-0.2838;0]$m,$\mathbf{v}_{40}=[0;-0.01;0]$m /s
The final state of the multi-spacecraft system in frame$\{\mathscr{B}\}$	S_1 $\boldsymbol{\rho}_B^{1c}=[0.4;1.2583;0]$m,$\mathbf{v}_B^{1c}=\mathbf{0}_{3\times1}$ S_2 $\boldsymbol{\rho}_B^{2c}=[-0.4;-1.2583;0]$m,$\mathbf{v}_B^{2c}=\mathbf{0}_{3\times1}$ S_3 $\boldsymbol{\rho}_B^{3c}=[-1.2583;0.4;0]$m,$\mathbf{v}_B^{3c}=\mathbf{0}_{3\times1}$ S_4 $\boldsymbol{\rho}_B^{4c}=[1.2583;-0.4;0]$m,$\mathbf{v}_B^{4c}=\mathbf{0}_{3\times1}$
The final state of the multi-spacecraft system in frame $\{\mathscr{L}\}$	$\boldsymbol{\rho}_{if}=\mathbf{T}_B^L\boldsymbol{\rho}_B^{ic}$,$\mathbf{v}_{if}=\boldsymbol{\omega}\times(\mathbf{T}_B^L\boldsymbol{\rho}_B^{ic})$
Definitional domain of PDE	T=200 s, $s_{\max}=5\times10^5$
Parameters of the metric matrix $\mathbf{G}$	$K_1=K_2=6\times10^3$, $\eta_1=\eta_2=0.02$

6.1. Comparative analysis

MPC-SCP is a commonly used algorithm in multi-spacecraft system reconfiguration problems [16]. It is capable of optimizing energy while achieving reconfiguration of the multi-spacecraft system configuration. The key aspect of this method is SCP, which is capable of solving optimal control problems.

Since the PDE provides a trajectory planning solution, a controller is needed to track and control the PDE solution from Algorithm 2. In this paper, a model predictive controller is selected for tracking. It is used to compare the energy consumption differences between the SCP and the manifold-based method in solving the optimal control problem.

In the simulation scenario, the modular multi-spacecraft system acquires the initial maneuvering velocity from the assembled state and makes a safe proximity to the tumbling target. The scenario parameters are shown in Table 1.

The results of the MPC-SCP and manifold-based method are shown in Fig. 10. Both methods allow multi-spacecraft system to be disassembled from the assembled state and arrive at the target position.

Safety is further analyzed. According to Table 1, the minimum collision avoidance radius in Eqs.(16) and (17) are $R_c=0.5657$mand $R_t=1.1340$m, respectively. The variation of distance between spacecrafts within the multi-spacecraft system for both methods is shown in Fig. 11 ((i,j) denotes the distance between$s_i$and s_j). The variation of the distance between the multi-spacecraft system and the target is shown in Fig. 12. It can be seen that the PDE can iterate the multi-spacecraft system trajectory toward the solution satisfying the collision avoidance constraints during the solution process. In addition, the time to approach the minimum collision avoidance radius is shorter for the

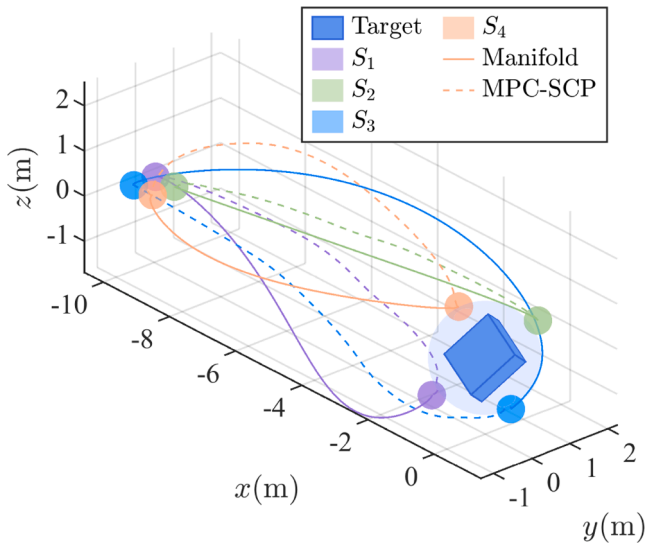
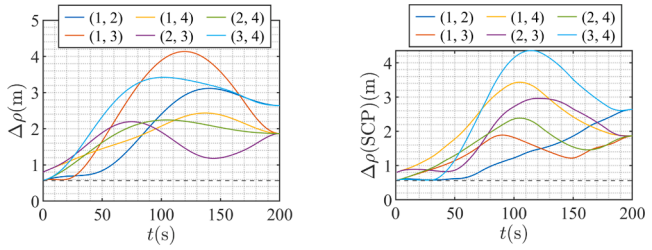


Fig. 10. Comparison of trajectory planning results.



(a) Manifold (b) MPC-SCP

Fig. 11. The distance change between spacecrafts within the multi-spacecraft system.

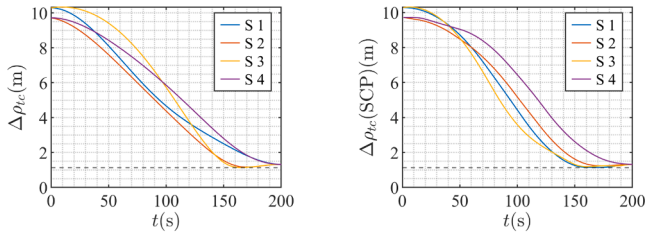
manifold-based method than for the MPC-SCP method. Therefore, the manifold-based method is safer, which is a result of using the potential function as a metric.

Comparison of velocity and control inputs for both methods are shown in Figs. 13-16. As can be seen in Fig. 13-14, the velocity planning results obtained using the manifold-based method are smoother. This is because the metric based on the potential function has good differentiability and clear physical meaning, which can ensure smooth velocity while handling collision avoidance constraints.

From Fig. 15-Fig. 16, it can be seen that the control inputs obtained by the manifold-based method are smoother. The results obtained by the MPC-SCP method are closer to an impulse form, with obvious jitter in the control curves. Further analysis of the variation of control inputs is shown in Fig. 15 and Fig. 16. The maximum control input ($\| \mathbf{u}_i \|_{\max}$), single spacecraft control input ($\sum_{t=0}^{T} \| \mathbf{u}_i(t) \|_2$), and multi-spacecraft system total control input ($\sum_{i=1}^{N} \sum_{t=0}^{T} \| \mathbf{u}_i(t) \|_2$) for both methods are compared as shown in the Table 2.

According to Table 2, the maximum control input and energy consumption of each module in the manifold-based method are one order of magnitude lower than those in the MPC-SCP method. Additionally, the total multi-spacecraft system energy consumption is two orders of magnitude smaller than that of the MPC-SCP method. This suggests that by characterizing the multi-spacecraft system state space within the manifold and designing the structure (metric) of this space in a principled manner, the manifold-based method is better equipped to avoid local optima.

The soft docking performance for safe proximity of rotating targets is further analyzed based on Fig. 2. The last 10 s before the end of the scene



(c) Manifold (d) MPC-SCP

Fig. 12. Distance change between multi-spacecraft system and target.

is taken for analysis. α is the angle between the instantaneous angular velocity $\boldsymbol{\omega}_c = \boldsymbol{\rho}_i(t) \times \mathbf{v}_i(t)$ of each mode in the multi-spacecraft system and the rotational velocity $\boldsymbol{\omega}$ of the target. The variation of α is shown in Fig. 17.

From Fig. 17, it is evident that the variation of α in the manifold-based method is significantly lower than that of the MPC-SCP method, particularly for module 1 (s_1) and module 3 (s_3). Minimizing the variation of α is crucial for enhancing soft docking performance. Consequently, the manifold-based method not only improves soft docking performance but also achieves fuel savings.

It is shown in Fig. 10 that the trajectories of s_1 and s_3 obtained by the two methods are reversed at the final approach moment. This is due to the fact that the soft docking performance is treated as a soft constraint for the MPC-SCP during the solution process. Whereas, the manifold-based method treats it as a hard constraint and gives it in the initial guess. Therefore, the manifold-based method can converge to a solution that is more consistent with the soft docking performance when searching for the optimal solution.

In order to evaluate the computational efficiency of the two methods, 30 trajectory planning sessions were performed for each method. The results are shown in Fig. 18. The average computation time of the MPC-SCP method is 29.2764 s, while the average time of the manifold-based method is significantly lower at 16.2254 s. Compared with MPC-SCP, the computational time using the manifold-based method is reduced by 44.58%, highlighting its superior computational efficiency.

6.2. The Impact of interpolation on algorithm performance

The linear interpolation in Eq. (30) is primarily achieved through temporal interpolation. The number of temporal interpolation points corresponds to the number of PDE grid points in Eq. (26). This section analyzes the impact of linear interpolation in Eq. (30) on trajectory smoothness and computational efficiency. Jerk directly reflects the smoothness of control inputs and serves as a key metric for evaluating trajectory smoothness. Let P_T denote the number of time interpolation points and $\Delta t = T/P_T$ denote the time step. The MAX Jerk (maximum jerk) and RMS Jerk (root-mean-square jerk) obtained from discrete-time simulations for each spacecraft are:

$$\text{MAXJerk} = \max_{i=1}^{P_T} \left\{ \sqrt{\left(\frac{\Delta u_x(i)}{\Delta t}\right)^2 + \left(\frac{\Delta u_y(i)}{\Delta t}\right)^2 + \left(\frac{\Delta u_z(i)}{\Delta t}\right)^2} \right\} \tag{35}$$

$$\text{RMSJerk} = \sqrt{\frac{1}{P_T} \sum_{i=1}^{P_T} \left[\left(\frac{\Delta u_x(i)}{\Delta t}\right)^2 + \left(\frac{\Delta u_y(i)}{\Delta t}\right)^2 + \left(\frac{\Delta u_z(i)}{\Delta t}\right)^2 \right]} \tag{36}$$

For five sets of time interpolation points with $P_T$values of 200, 250, 300, 350, and 400, the trends in MAX Jerk and RMS Jerk for the four spacecraft are shown in Fig. 19 and Fig. 20.

As shown in the Fig. 19 and Fig. 20, when P_T=200, the maximum MAX Jerk among the four spacecraft is 2.93×10^{-4}, and the maximum RMS Jerk is 2.34×10^{-5}, indicating good trajectory smoothness. Furthermore, as the number of interpolation points increases, both MAX

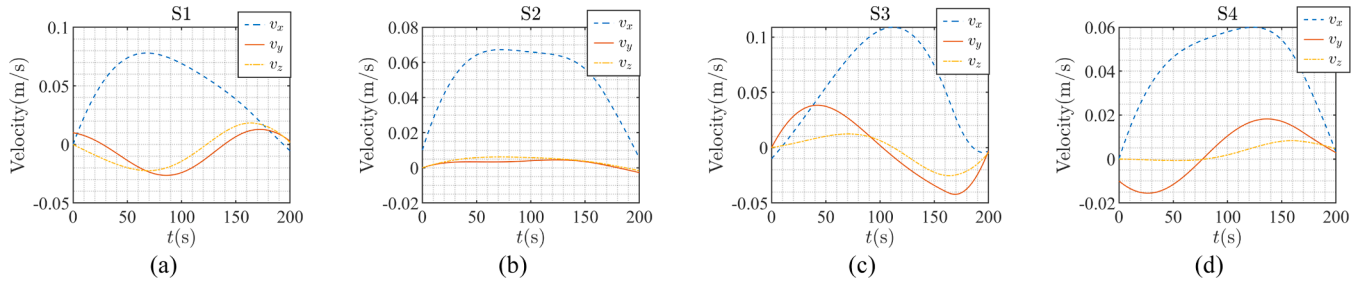


Fig. 13. multi-spacecraft system velocity changes (Manifold-based method).

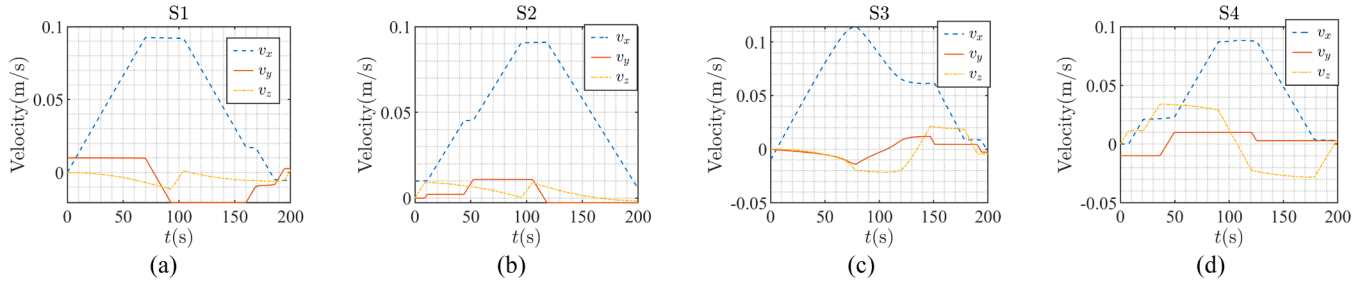


Fig. 14. multi-spacecraft system velocity changes (MPC-SCP method).

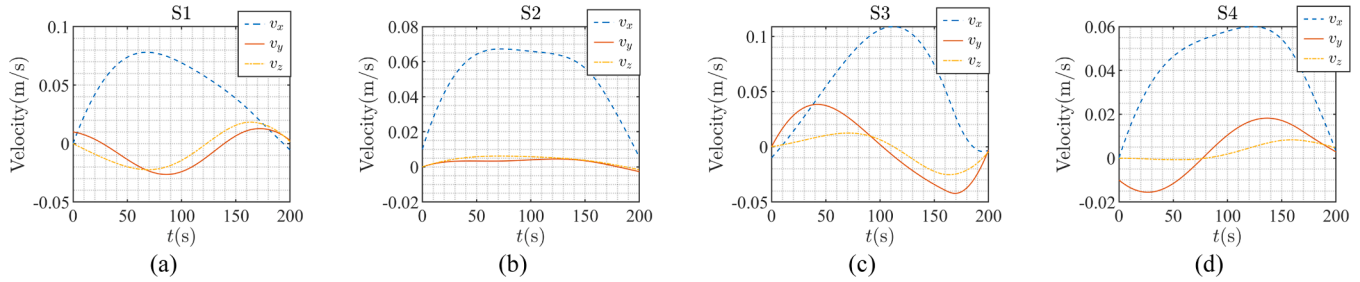


Fig. 15. Variation of multi-spacecraft system control inputs (Manifold-based method).

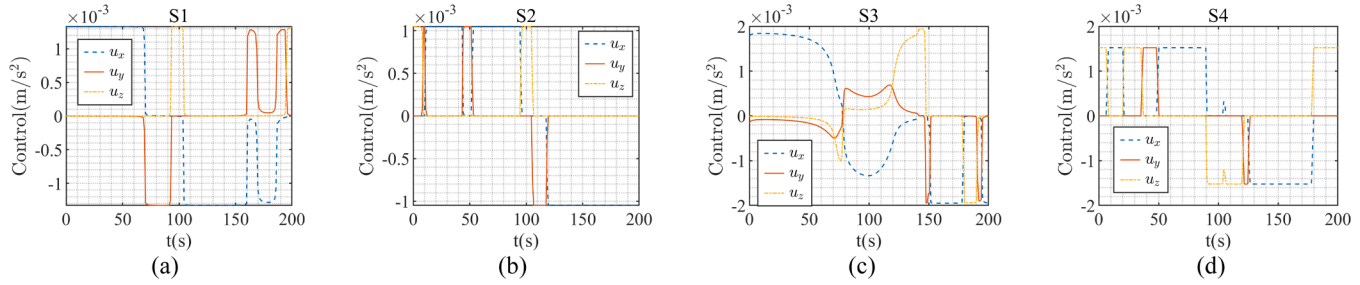
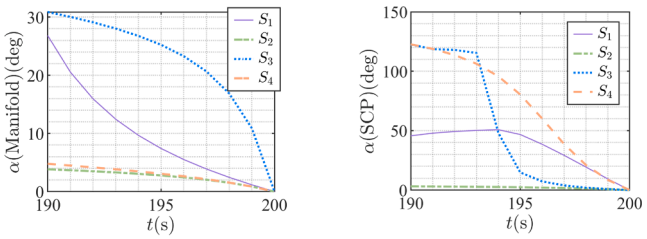


Fig. 16. Variation of multi-spacecraft system control inputs (MPC-SCP method).

Table 2
Comparison of Control Inputs.

Control inputs (m/s^2)	Manifold				MPC-SCP			
	s_1	s_2	s_3	s_4	s_1	s_2	s_3	s_4
$\|\| \mathbf{u}_i \|\|_{\max}$	2.884×10^{-4}	2.663×10^{-4}	2.963×10^{-4}	1.762×10^{-4}	1.331×10^{-3}	1.048×10^{-3}	1.952×10^{-3}	1.524×10^{-3}
$\sum_{t=0}^{T} \|\| \mathbf{u}_i(t) \|\|_2$	0.0139	0.0351	0.0164	0.0139	0.2618	0.2084	0.3301	0.3016
$\sum_{i=1}^{N} \sum_{t=0}^{T} \|\| \mathbf{u}_i(t) \|\|_2$	0.0893				1.1019			



(a) Manifold (b) MPC-SCP

Fig. 17. Variation of α.

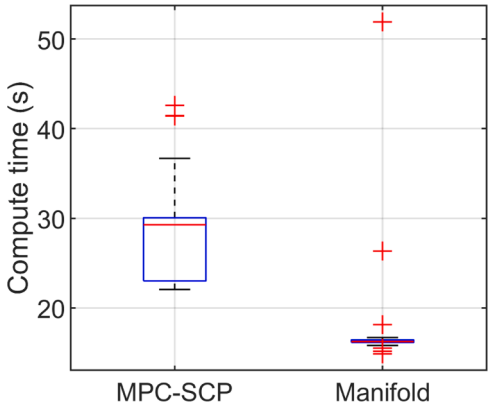


Fig. 18. Comparison of compute time.

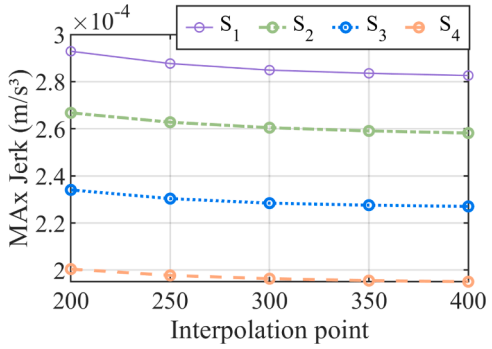


Fig. 19. Variation in MAX Jerk with the number of interpolation points.

Jerk and RMS Jerk exhibit a decreasing trend, leading to improved trajectory smoothness. However, increasing the number of interpolation points results in a denser PDE grid, thereby reducing computational efficiency. The computation times for the manifold-based method corresponding to five sets of different time interpolation point counts are shown in Fig. 21.

As shown in Fig. 21, compute time increases with the number of interpolation points. Therefore, the selection of interpolation points should balance trajectory smoothness and computational speed. Considering that P_T=200 yields satisfactory trajectory smoothness, P_T=200 is chosen as the time interpolation value.

Furthermore, from a computational perspective, the core of this algorithm lies in replacing the optimization process with solving a PDE. The PDE iteratively evolves the trajectory from an initial guess toward a solution that satisfies both dynamic and collision avoidance constraints while exhibiting soft docking performance. Consequently, linear interpolation primarily affects the mesh density of the PDE, impacting its computational speed. Since the final trajectory complies with dynamic

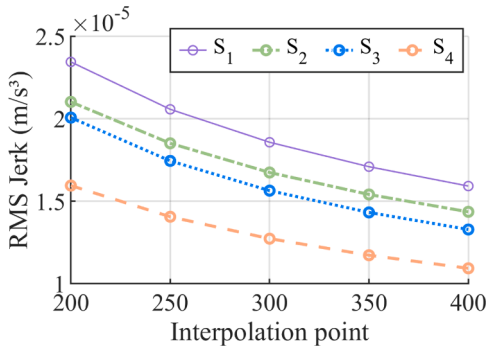


Fig. 20. Variation in RMS Jerk with the number of interpolation points.

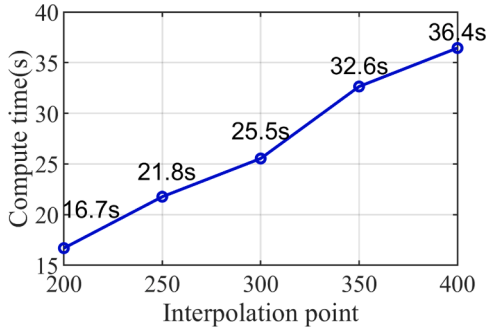


Fig. 21. Variation of compute time with the number of interpolation points.

requirements, the interpolation has a minimal effect on trajectory smoothness.

6.3. Homotopy iteration process analysis

The iterative process of solving the geodesic by the manifold-based method is analyzed. As shown in Fig. 22, the PDE is able to iterate from the initial guess of the homotopy based on the rotated 3D-Dubins curves of Section 4. The corresponding trajectories form the visualization iteration space.

In addition, the PDE allows the velocity to be iterated from Eq. (31) to a result that satisfies the dynamics. Given that velocity vectors in space are three-dimensional, to visualize this iterative process, the velocity mode $\| \mathbf{v}(t) \|_2$ of the module at time t is computed. The results are presented in Fig. 23. The two dimensions of the PDE are t and s. Seven homotopy data slices are taken within the interval $s \in [0, s_{\max}]$. It can be observed that the slice at s=0 represents a linearly varying initial guess as described by Eq. (31). As the iteration progresses from s=0 to $s = s_{\max}$, the velocity modes of the four module spacecraft exhibit a gradual change over time, eventually converging to the slicing result at $s = s_{\max}$.

Finally, the energy during the PDE iterations is analyzed. The sum $\sum_{t=0}^{T} \| \mathbf{v}_i(t) \|_2$ of the velocity modes of the four modular spacecraft in time T is used as the energy metric. As the homotopy number changes, the energy of the four spacecraft changes as shown in Fig. 24. It can be seen that the PDE evolves in the direction of decreasing energy during the homotopy iteration process.

6.4. Experiments on air-floating platform

In order to further validate the effectiveness of the proposed methodology, experiments were conducted using the air flotation platform experimental system detailed in references [36] and [37]. The experimental system consists of a target and four simulators. The target spins around the z-axis with an angular velocity of $\omega = -0.5\mathrm{r}/\mathrm{min}$ and has a shape envelope of $0.66 \times 0.66 \times 0.6\ \mathrm{m}^3$. The simulator has a maximum

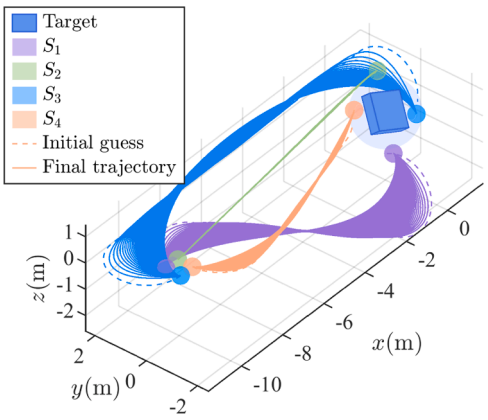


Fig. 22. Iterative space of trajectories for PDE.

thrust of 2.1 N in a single direction and is equipped with an integrated computer. The processor is an Intel J1800 CPU running at 2.41 GHz. The height difference at the joint of the marble platform is less than 10 μm, and the horizontality error is less than 10 μm. The simulators have external dimensions of cylinders with a radius of 0.25 m and a height of 0.6 m.

It should be specifically noted that since this air flotation platform primarily simulates microgravity conditions within a two-dimensional plane through jet thrust, current experimental verification focuses primarily on target motion within this plane. Although this represents a simplified motion model, it constitutes a typical and important subcase of three-dimensional tumbling motion in terms of dynamics. The scenario parameters are set as shown in Table 3.

The platform exhibits inherent initial positioning errors, primarily originating from three sources: (1) uneven air-floating forces and manual disturbances during initialization, (2) joint gaps and surface flatness imperfections of the marble platform, and (3) deviations in electromagnetic valve thrust and control algorithms. To account for these uncertainties, three independent experimental trials were conducted. A video screenshot of one set of experiments is shown in Fig. 25. The target undergoes a spinning motion, and the simulators separate from an initial proximity state to reach their final positions. Finally, the simulators track the target's rotation.

The experimental data processing results are shown in Fig. 26. It can be seen from Fig. 26(a) that there are errors in the initial positioning of the three groups of experiments, but the simulators can achieve safe proximity to the rotating target. Fig. 26(d)-(e) show that the manifold-based method can achieve a circumferential spiral approach to ensure soft docking performance.

The distances between the simulators and between them and the target are shown in Fig. 27 and Fig. 28. The distance between the simulators is greater than $R_c = 0.5$m, and the distance between the simulators and the target is greater than $R_t = 0.7167$m, ensuring safety during the approach process.

Further analysis of the algorithm's traceability is conducted. The root mean square error (RMSE), mean absolute error (MAE), and maximum error between the experimental trajectory and the theoretical trajectory are shown in Table 4.

Data analysis in Table 4 indicates that the proposed algorithm achieves centimeter-level accuracy in tracking control. The average RMSE is 1.66 cm, the average MAE is 1.42 cm, and the average maximum error is 3.41 cm. Although brief performance fluctuations occurred in individual

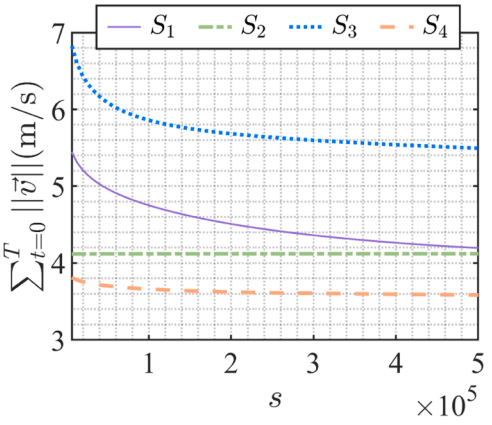


Fig. 24. Multi-spacecraft system energy evolution during the PDE solving process.

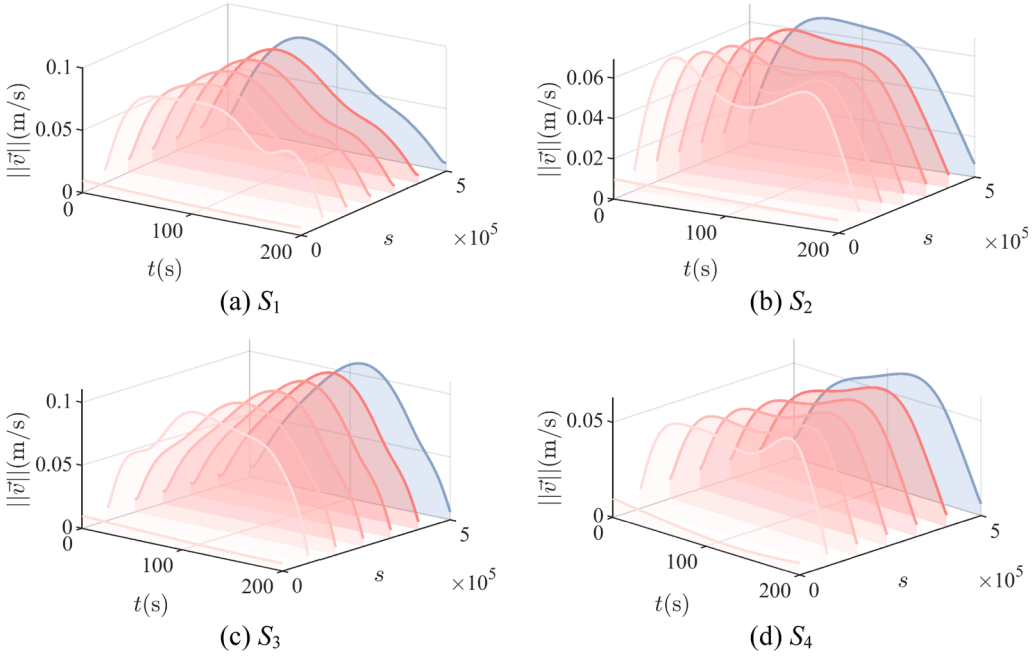


Fig. 23. Iterative visualization of velocity modes of modular spacecraft during PDE solution processes.

Table 3
Experimental scenario parameters.

initial state	Final state (in frame $\{\mathscr{B}\}$)	Other parameters
$\boldsymbol{\rho}_{10} = [-1.85; 2.35; 0]\text{m}, \mathbf{v}_{10} = [0; 0; 0]\text{m/s}$	$\boldsymbol{\rho}_{1f} = [0; 0.9167; 0]\text{m}, \mathbf{v}_{1f} = [0; 0; 0]\text{m/s}$	Target position $\boldsymbol{\rho}_t = [1.5; 2; 0]\text{m}$
$\boldsymbol{\rho}_{20} = [-1.15; 2.35; 0]\text{m}, \mathbf{v}_{20} = [0; 0; 0]\text{m/s}$	$\boldsymbol{\rho}_{2f} = [0; -0.9167; 0]\text{m}, \mathbf{v}_{2f} = [0; 0; 0]\text{m/s}$	Scenario duration $T = 100s$
$\boldsymbol{\rho}_{30} = [-1.85; 1.65; 0]\text{m}, \mathbf{v}_{30} = [0; 0; 0]\text{m/s}$	$\boldsymbol{\rho}_{3f} = [-0.9167; 0; 0]\text{m}, \mathbf{v}_{3f} = [0; 0; 0]\text{m/s}$	$R_c = 0.5\text{m}$
$\boldsymbol{\rho}_{40} = [-1.15; 1.65; 0]\text{m}, \mathbf{v}_{40} = [0; 0; 0]\text{m/s}$	$\boldsymbol{\rho}_{4f} = [0.9167; 0; 0]\text{m}, \mathbf{v}_{4f} = [0; 0; 0]\text{m/s}$	$R_t = 0.7167\text{m}$

tests, the algorithm demonstrated overall stability. The maximum error, which remained within acceptable limits, primarily occurred within the first 26 seconds of the experiment and was attributed to initial positioning errors.

The experimental results show that the trajectory generated by the manifold method still has good traceability in the presence of errors and disturbances. However, the actual environment of spacecraft in orbit involves complex nonlinear dynamics and non-negligible external disturbances [38,39]. Therefore, based on the existing manifold trajectory planning framework, subsequent research needs to further incorporate nonlinear dynamic models and uncertainties, and enhance the robustness and applicability of the manifold method by constructing adaptive metrics.

7. Conclusions

In this paper, a trajectory planning method based on manifold is proposed for the safe approach trajectory planning of the module multi-spacecraft system to the tumbling target. By comparing with the model predictive control-sequential convex programming (MPC-SCP) method, the effectiveness is verified by the air flotation platform. The simulation results demonstrate that, in comparison to the MPC-SCP method, the manifold-based approach not only reduces fuel consumption but also exhibits superior soft docking performance. Additionally, its computational efficiency is enhanced by 44.58%. This is because, first of all, the manifold-based method can transform the optimal control problem under multiple constraints into a partial differential equation (PDE) solving problem, which reduces the dependence on the initial conjecture. Secondly, the homotopy class construction method based on the

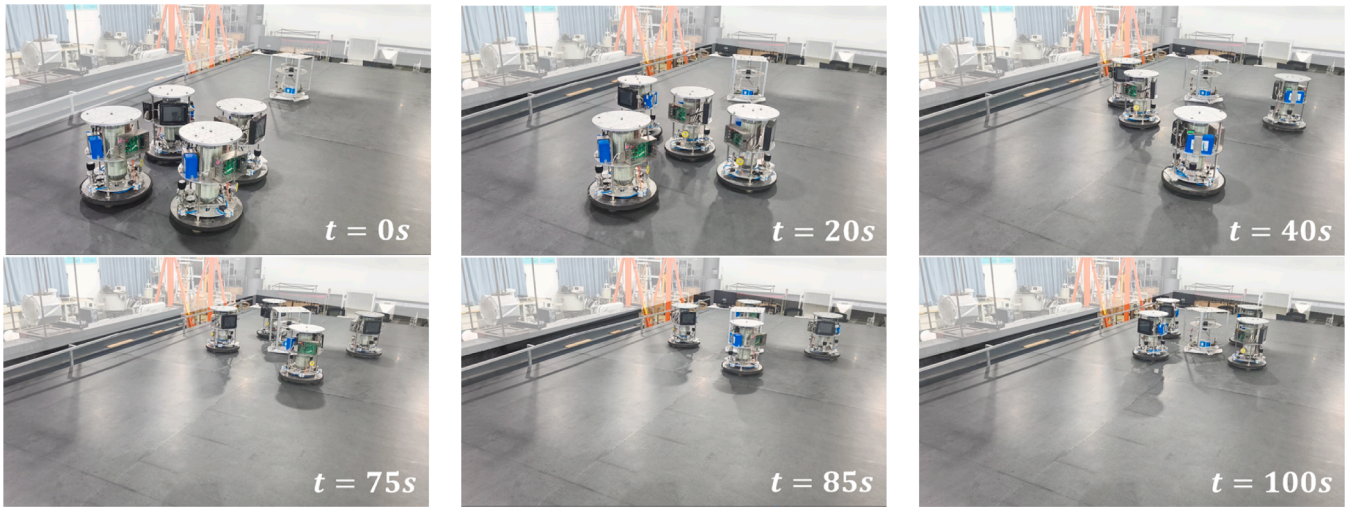


Fig. 25. Screenshots of the experimental process.

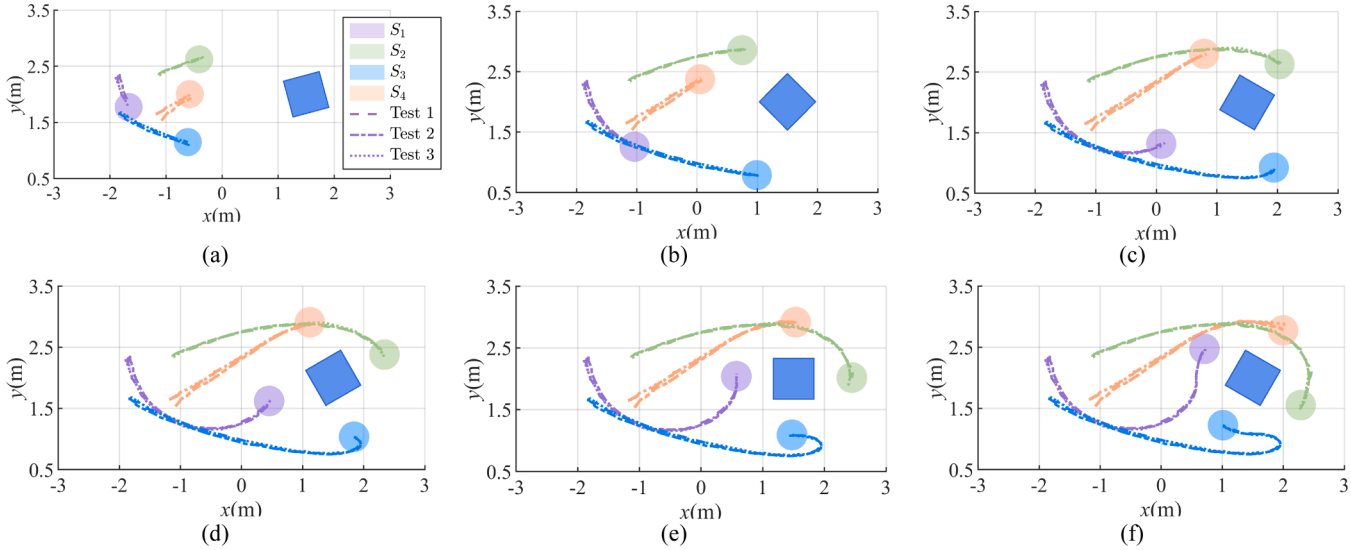


Fig. 26. The trajectories of the simulators.

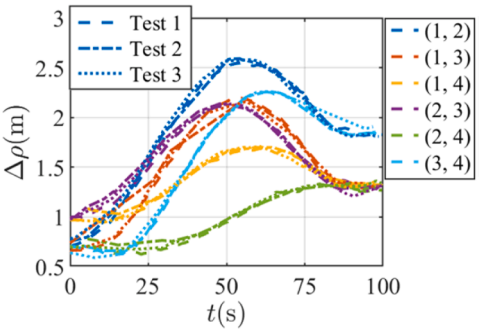


Fig. 27. Distance between the simulators.

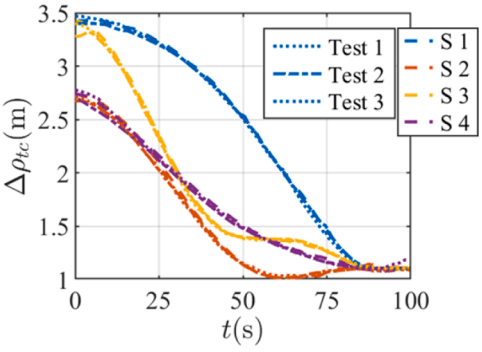


Fig. 28. Distance between the simulators and the target.

Table 4
Tracking error analysis results.

Test Number	Spacecraft	RMSE(m)	MAE(m)	Max Error(m)
Test 1	S_1	0.0131	0.0115	0.0211
	S_2	0.0144	0.0121	0.0305
	S_3	0.0085	0.0076	0.0150
	S_4	0.0105	0.0091	0.0214
Test 2	S_1	0.0211	0.0196	0.0319
	S_2	0.0109	0.0100	0.0181
	S_3	0.0231	0.0186	0.0434
	S_4	0.0525	0.0400	0.1289
Test 3	S_1	0.0221	0.0176	0.0528
	S_2	0.0138	0.0109	0.0281
	S_3	0.0209	0.0157	0.0527
	S_4	0.0113	0.0096	0.0221

rotated 3D-Dubins curve can make the PDE converge from the initial guess that does not satisfy the dynamic constraints while considering the soft docking. Finally, based on the air flotation experimental system, the semi-physical simulation experiment is carried out to verify the effectiveness of the manifold-based method.

CRediT authorship contribution statement

Jingxian Wang: Writing – original draft, Validation, Methodology, Conceptualization. **Rong Chen:** Writing – review & editing, Validation, Data curation. **Heng Zhou:** Investigation, Formal analysis. **Chenyu Shan:** Validation, Investigation. **Yong Zhao:** Supervision, Project administration, Funding acquisition. **Yuzhu Bai:** Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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